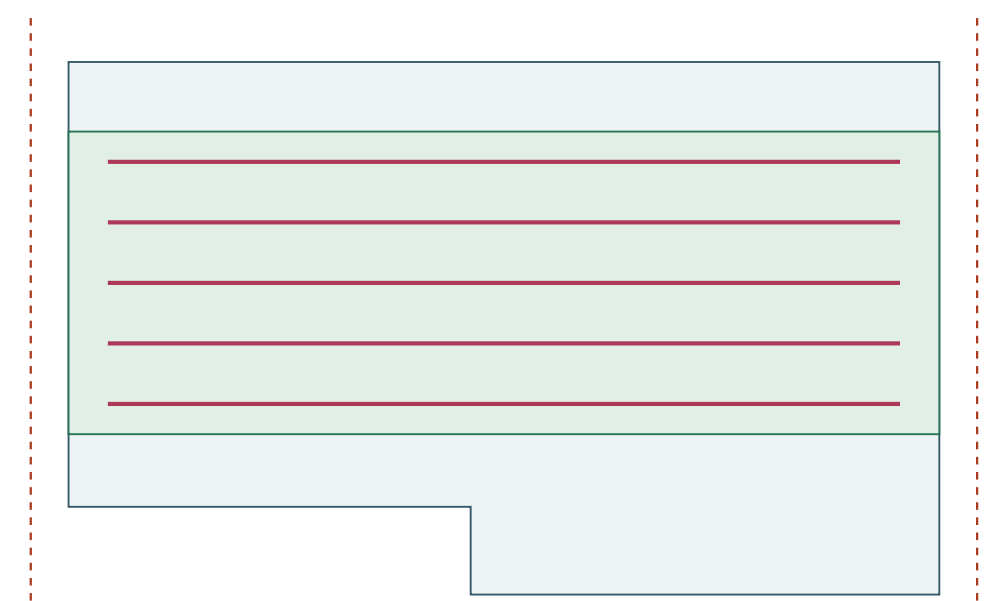


RC-001 Stair landing: loads to a reinforcement sketch

SIMULATED COORDINATION REVIEW CASE STUDY | NOT FOR CONSTRUCTION | 2026-09-13

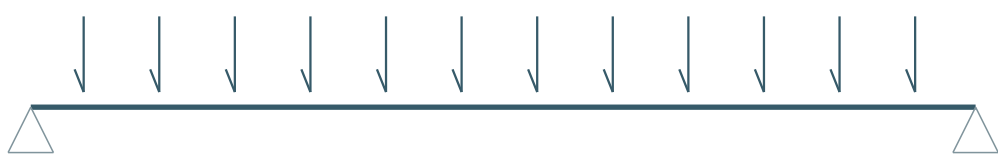
LOCAL GRAVITY STRIP ONLY | no whole-landing, anchorage or seismic acceptance

1 LOCATE THE STRIP / PLAN / 1:25 AT A3



Green: y = 10.20..11.20 m; main bottom bars run along X.
Slab top +2.00 m; h = 180 mm; footprint follows the source IFC.
Band excludes flight bearings; adjacent plate action is not checked.

2 RELEASED LOCAL MODEL / DIAGRAMMATIC

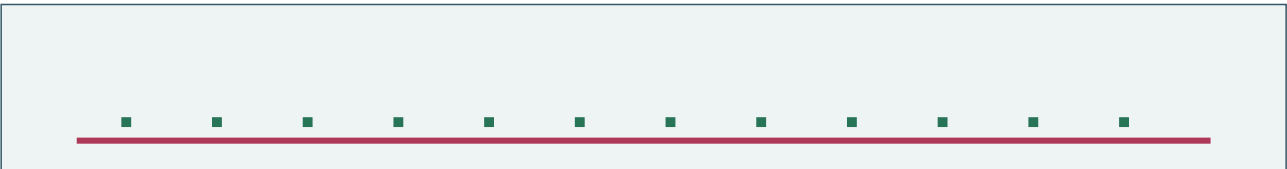


Support-line span 3.13 m; loaded length 2.88 m
1.2D + 1.6L: q = 11.691 kN/m

Inherited nominal wall reaction lines x = 6.245 / 9.375 m.
Pin/roller idealization; unloaded 0.125 m end zones.
No new gravity is added to the whole-building model.

SOURCE GUID: 3WBnbFw6v18ePpRsZLm4Gj | Article illustration within A01; NOT a complete stair reinforcement sheet.

3 SECTION AND SELECTED CHECKS



Midspan section sketch; h 180 / cover 20 mm (not to scale)
Magenta: bottom main D10 @ 200; green: transverse above.

Concrete f_c = 28 MPa; steel f_y = 420 MPa (case assumption)
Effective depth d = 155.23 mm
Demand M_u = 14.23 kNm/m
Provided ϕM_n = 20.50 kNm/m
 A_s = 356.65 $\geq$ 324 mm²/m minimum
 V_u = 16.84 $\leq$ ϕV_c = 55.28 kN/m
Selected gravity strength, minimum steel and spacing: PASS
D10 @ 250 fails minimum steel; D13 @ 300 uses more steel.

SUPPORT / END DETAILS: HOLD
Anchorage, continuity, integrity steel and cut lengths are unresolved.
Lines stop schematically; they are NOT proposed bar terminations.
D10 @ 200 also used for transverse distribution in this strip.

Taiwan concrete code effective 2024-01-01, with 2024 errata.
Tables 5.3.1 / 21.2.2 / 22.5.5.1; sections 7.6 / 7.7 / 24 / 25.
Read landing-strip.md / .json for the exact scope, inputs and checks.